

Multi-Table Sales Analysis

Northwind Trading Dataset: SQL Analysis Report

Veda Technology | Data Analytics Track | Level 2, Task 25

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Total Revenue	Total Orders	Customers	Avg Order Value
Rs. 12.03 Cr	830	89	Rs. 1,44,880

1. Objective and Approach

The goal of this task was to combine the orders, products and customers tables of the Northwind dataset and apply relational analysis end to end, producing a SQL script, supporting visuals and five business insights.

Tools used: SQL (SQLite in Google Colab), Python (pandas, matplotlib).

Dataset

The Northwind sample database covers 830 orders placed by 89 customers between 04 Jul 1996 and 06 May 1998. Tables used: orders, order_details, products, categories, customers, employees, suppliers and shippers. They were joined into one analysis table at order-line level (2,155 rows).

Revenue definition

Revenue is calculated after discount: **unitPrice x quantity x (1 - discount)**. The original Northwind amounts are in US dollars and have been converted to Indian rupees at an assumed rate of **Rs. 95 per USD** (approximate rate, October 2026). In the charts, values are shown in lakh (1 lakh = 1,00,000); Rs. 1 crore = 100 lakh. Orders are counted with COUNT(DISTINCT orderID) because one order contains multiple product lines.

Join Validation

Joins were validated by comparing row counts and totals before and after merging, to make sure no rows were duplicated or lost.

Check	Expected	Result
order_details rows	2,155	2,155
Distinct orders	830	830
Total revenue (after discount, Rs.)	12,02,50,336.90	12,02,50,336.90
Rows with missing product/customer	0	0

Double counting was avoided in two ways: orders were counted with DISTINCT, and freight (an order-level value) was never summed across order lines.

2. SQL Script

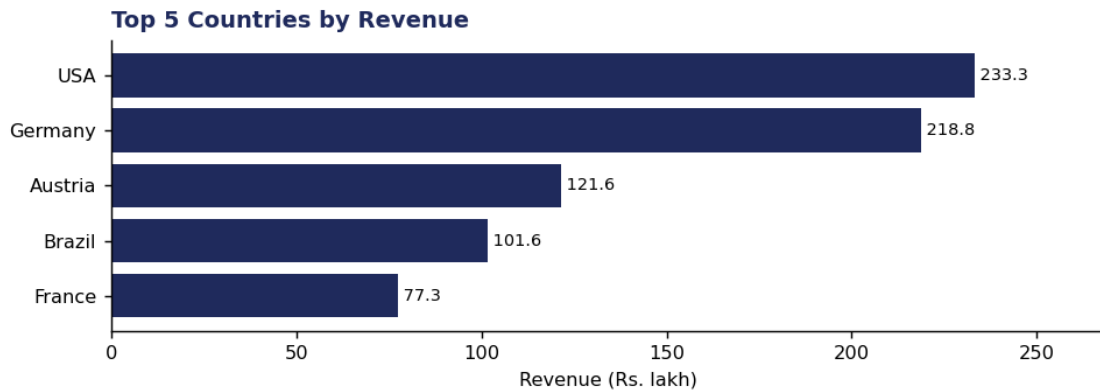
Main multi-table query (country and category sales). All five insight queries follow the same JOIN pattern.

```
SELECT c.country, cat.categoryName,
       COUNT(DISTINCT o.orderID) AS total_orders,
       ROUND(SUM(od.unitPrice * od.quantity * (1 - od.discount)), 2) AS revenue
FROM orders o
JOIN customers c ON o.customerID = c.customerID
JOIN order_details od ON o.orderID = od.orderID
JOIN products p ON od.productID = p.productID
JOIN categories cat ON p.categoryID = cat.categoryID
GROUP BY c.country, cat.categoryName
ORDER BY revenue DESC;
```

3. Key Insights

Insight 1: USA and Germany drive the business

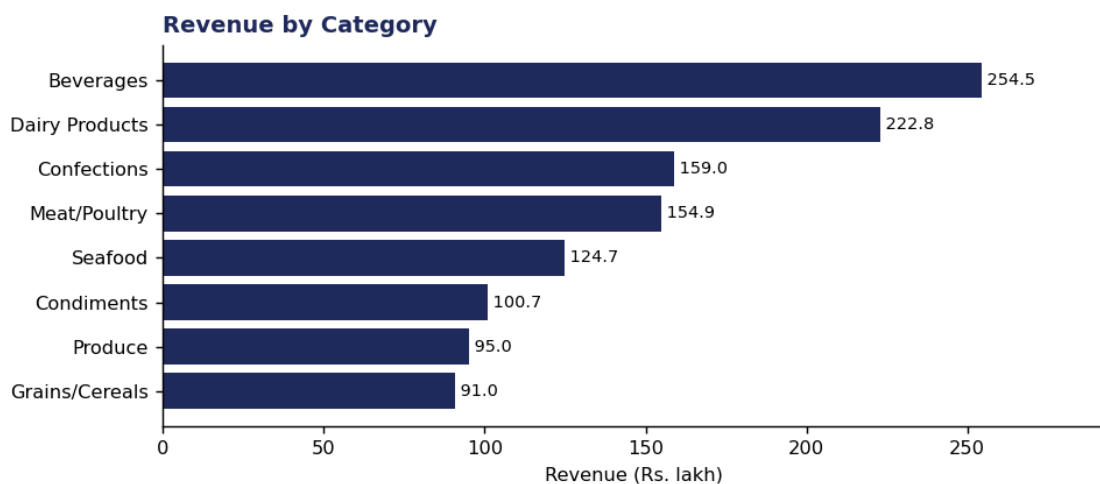
USA (Rs. 2,33,30,539) and Germany (Rs. 2,18,77,040) together generate 38% of total revenue. Both have 122 orders, but the USA has the higher revenue. Austria ranks third with only 40 orders, which shows a very high value per order (about Rs. 3,04,009).



Country	Orders	Revenue (Rs.)
USA	122	2,33,30,539
Germany	122	2,18,77,040
Austria	40	1,21,60,366
Brazil	83	1,01,57,949
France	77	77,29,040

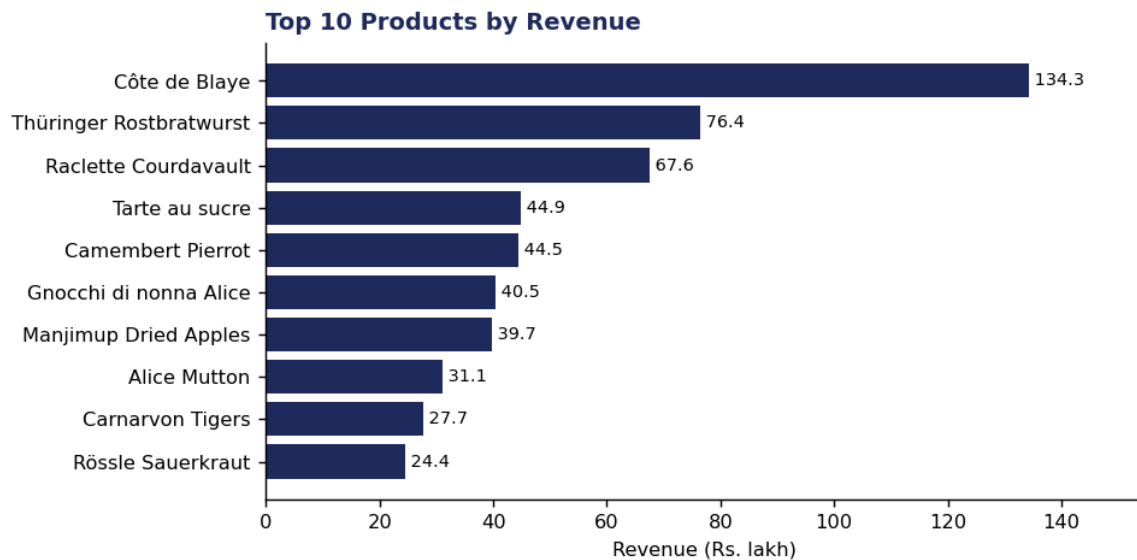
Insight 2: Beverages and Dairy lead the categories

Beverages (Rs. 2,54,47,477) and Dairy Products (Rs. 2,22,78,193) are the two largest categories and make up 40% of revenue. Grains/Cereals and Produce are the weakest.



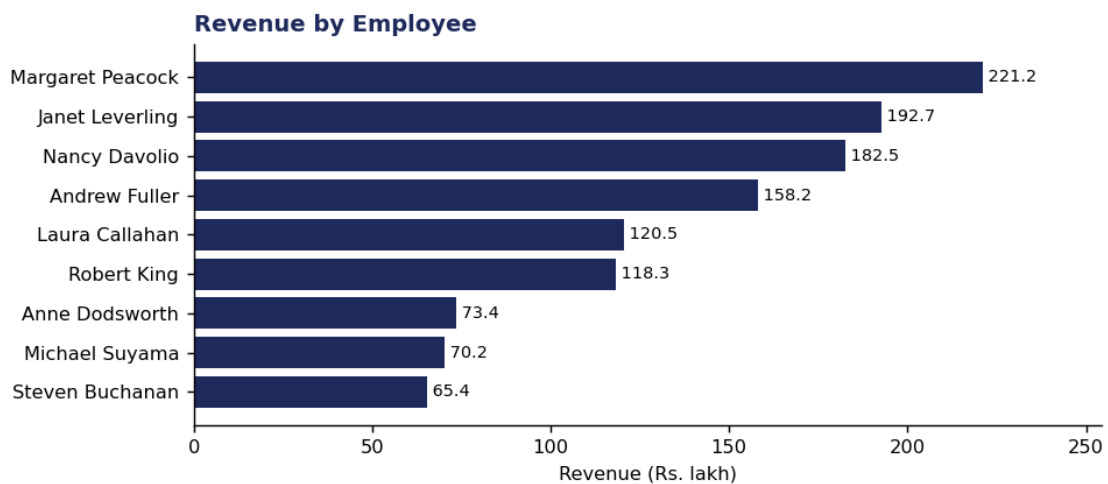
Insight 3: One product, Cote de Blaye, is a major revenue source

Cote de Blaye alone earns Rs. 1,34,32,690 (11.2% of all revenue) from just 24 orders, because of its very high unit price. Thüringer Rostbratwurst is second at Rs. 76,35,023. The top 10 products together account for 44% of revenue, so stock and pricing of these items deserve close attention.



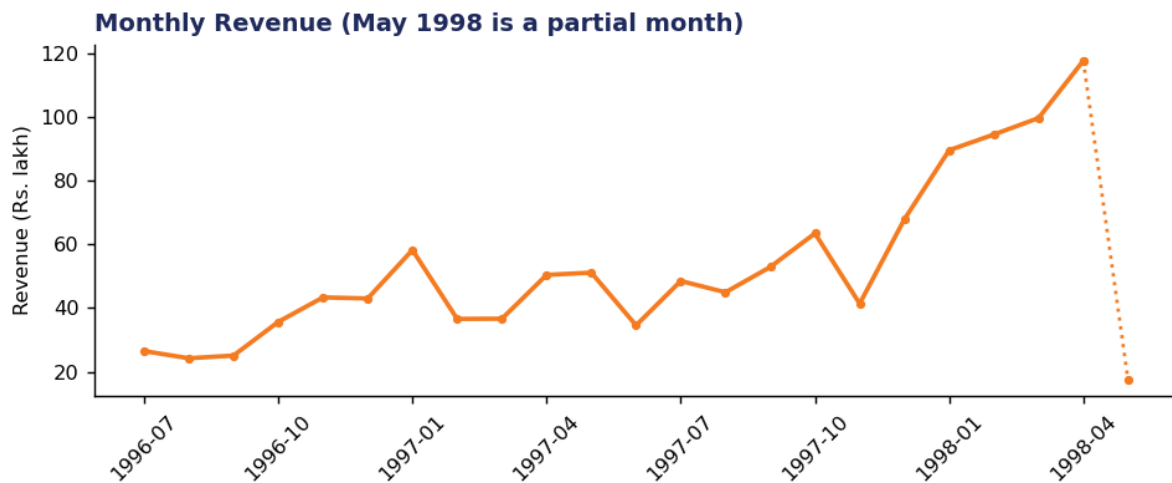
Insight 4: Margaret Peacock is the top-performing employee

Margaret Peacock handled 156 orders worth Rs. 2,21,24,633, the highest of all employees. Janet Leverling and Nancy Davolio follow. Anne Dodsworth, Steven Buchanan and Michael Suyama sit at the bottom of the ranking.



Insight 5: Revenue grew strongly through 1997 and into 1998

Annual revenue rose from Rs. 1,97,67,977 in 1996 (July to December only) to Rs. 5,86,23,097 in 1997. 1998 data covers only January to early May, yet it already reached Rs. 4,18,59,263. A like-for-like comparison of January to April shows revenue up **121%** in 1998 (Rs. 4,01,17,569) versus 1997 (Rs. 1,81,75,579). April 1998 was the best month overall at Rs. 1,17,60,874. May 1998 looks low only because the data ends on 6 May.



4. Top Customers

Customer	Orders	Revenue (Rs.)
QUICK-Stop	28	1,04,76,345
Ernst Handel	30	99,63,124
Save-a-lot Markets	31	99,14,386
Rattlesnake Canyon Grocery	18	48,54,290
Hungry Owl All-Night Grocers	19	47,48,091

QUICK-Stop, Ernst Handel and Save-a-lot Markets each bring in more than Rs. 95 lakh, roughly twice the next customer.

5. Conclusion and Recommendations

- Focus sales and marketing on the USA and Germany, and study why Austria has such high value per order.
- Protect supply of Beverages and Dairy, the two categories that earn the most.
- Reduce dependence on a single high-priced product (Cote de Blaye) by promoting other top sellers.
- Share the practices of top employees with lower-ranked ones to lift overall performance.
- Maintain the 1998 growth momentum by retaining the largest customers.